# Exact Newform-Period Taxonomy on a Frozen Finite Hecke-Output Multiset for a Level-11 Time Change of the Modular Geodesic Flow

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**Abstract**

We study a positive time change of the geodesic flow on the level-11 modular surface whose first period variation is the real period of the unique normalized weight-two newform. The motivating question is whether the Hecke action on these periods can induce a primitive-orbit Euler recurrence. We first identify the correct owner of the Hecke relation: the pushforward of an oriented closed cycle, which is generally a sum of closed geodesic owners rather than a single primitive orbit. In the canonical inverse-closed orbit product, odd variations cancel. At second order, an all-parameter scalar recurrence is equivalent to exact degree-wise quadratic moment conditions. We then give a computer-assisted but exact classification of the frozen finite output multiset. A rational Schreier model of $H_1(Y_0(11), \mathbb{Q})$ has coordinates $(x, y, z)$, real involution $\tau(x, y, z) = (-x, y + z, -z)$, and real-period coordinate $k = 2y + z$. This makes every normalized period ratio rational and every quadratic moment a rational sum of squares. Among 138 Hecke cycle-owner instances, exactly two are full complex-period kernels, two are real-projection-only kernels, and 134 are true nonkernels. The candidate laws $\lambda_p = a_p$ and $\lambda_p = a_p^2$ each fail exactly 51 of 55 frozen groups; the control $a_p^2 - p$ fails all 55. These results refute the tested naive Hecke-like recurrences on the registered finite ledger, but they do not establish a global determinant, a prime–orbit correspondence, or a Route-A dynamical-zeta match.

**Keywords:** modular geodesic flow; newform periods; Hecke correspondence; time change; Schreier homology; dynamical zeta obstruction

## 繁體中文摘要

本文研究模曲面 $Y_0(11)$ 測地流的一族正時間變換，其閉軌道週期的一階變分由唯一正規化權二新形式之實週期給出。核心問題是：Hecke 對應在線性週期上的特徵關係，能否提升為逐一原始軌道的 Euler 型遞推。研究首先證明，正確的 Hecke 所有者是有向閉圈的推前，通常分解為多個閉測地線，而非單一原始軌道。標準的逆向封閉乘積使奇數階變分成對消失；二階恆等式則等價於按 Hecke 圈長分組的精確平方矩條件。本文以有理 Schreier 同調模型完成有限分類：$H_1(Y_0(11), \mathbb{Q})$ 中的實結構為 $\tau(x, y, z) = (-x, y + z, -z)$，實週期只取決於 $k = 2y + z$。因此所有正規化週期比與平方矩皆為有理數，可完全排除浮點零判定。凍結資料中的一百三十八個 Hecke 圈所有者，恰有兩個完整複週期核、兩個僅實投影核及一百三十四個真非核；候選律 $a_p$ 與 $a_p^2$ 各在五十五組中精確失敗五十一組，控制律 $a_p^2 - p$ 則全數失敗。此結果是有限輸出多重集合上的精確否定定理，不代表完整原始軌道普查，也不建立質數對應、全域動力行列式或 Hilbert–Pólya 實現。

關鍵詞： 模測地流；新形式週期；Hecke 對應；時間變換；Schreier 同調；動力 zeta 障礙

# 1 Introduction

Arithmetic surfaces offer a natural place to ask how algebraic correspondences interact with periodic-orbit expansions. On the level-11 modular curve, the unique normalized weight-two newform supplies a canonical holomorphic differential, and its real part can be used as the infinitesimal clock change of the geodesic flow. The resulting period variation is intrinsic: it is an integral over an oriented closed geodesic, is invariant under subgroup conjugacy, changes sign under orientation reversal, and scales linearly under traversal repetition. These properties make the construction a useful test bed for the stronger idea that Hecke structure might descend to a primitive dynamical Euler product.

The central research question is deliberately narrower than that broad motivation:

> For the frozen level-11 newform time change, do either of the scalar, all-$s$ second-variation recurrences $\lambda_p = a_p$ or $\lambda_p = a_p^2$ hold on the 55 registered source-word/prime groups formed from the frozen 138 correspondence-component owner instances, each counted with its declared Hecke-output multiplicity?

The answer is no for 51 of the 55 registered groups under each primary law; exactly four kernel groups pass, while the predeclared secondary control $a_p^2 - p$ fails all 55. This is a statement about the registered correspondence-component multiset with its declared finite multiplicity, not about a canonical global primitive-owner set. The Hecke correspondence acts on a cycle by producing a finite sum of closed owners of different branch-cycle degrees. Those degrees alter the length kernels in the formal expansion and are not zeta repetitions. At second order, inverse orientations add rather than cancel, and the resulting sum depends on squares of owner periods, so a scalar recurrence imposes degree-wise quadratic moments not implied by the linear Hecke eigenperiod relation.

A target-blind exact audit using the matched functionals $y - z$ and $y - 2z$ further separates the finite obstruction from the chosen newform coordinate $k = 2y + z$. Among the failures for $k$, both controls fail in 51, 44, and 55 groups for $a_p$, $a_p^2$, and $a_p^2 - p$, respectively; the seven remaining $a_p^2$ failures fail exactly one control, and no $k$-failure is accompanied by passes for both controls. Thus the added evidence supports a generic finite obstruction relative to this predeclared two-control panel. It is neither a universal theorem for all closed one-forms nor a newform-uniqueness result.

This article makes four contributions. First, it fixes the positive time change and proves the conjugacy, orientation, and repetition laws of its period coordinate. Second, it states the cycle-pushforward Hecke theorem with an explicit closed-owner decomposition, separating a Hecke branch-cycle degree from a zeta repetition index. Third, it derives exact first- and second-variation formulas and proves the necessary-and-sufficient quadratic degree-moment criterion. Fourth, it closes the registered finite classification with exact rational homology: all 138 frozen correspondence-component instances and all 55 source-word/prime groups are classified without a floating-point zero test, and a target-blind two-functional control decomposition shows that none of the studied-coordinate failures leaves a residue on which both matched controls pass. The last statement is bounded to that control panel and does not establish newform uniqueness.

The conclusion is an obstruction, not a positive determinant construction. The exact taxonomy shows that 134 output instances carry genuine nonzero real-period mass, while the only four kernel instances are explained by compact-homology vanishing or real-structure parity. The result neither enumerates every primitive $\Gamma_0(11)$ class nor defines a globally continued dynamical zeta function. It also does not use rational-prime targets or Riemann-zero data. These boundaries are part of the theorem statement rather than qualifications added after computation.

## 2 Mathematical setting

### 2.1 The level-11 differential and time change

Let

$$Y_0(11) = \Gamma_0(11)\backslash\mathbb{H}, \qquad f(z) = \eta(z)^2\eta(11z)^2 = \sum_{n\geq 1} a_n q^n, \quad q = e^{2\pi i z}.$$

The form $f$ is the normalized weight-two newform of level 11 with trivial character. The corresponding newspace has dimension one, the coefficient field is $\mathbb{Q}$, and the displayed eta product has $a_1 = 1$. [4, Properties and q-expansion] Define

$$\omega_f = 2\pi i f(z)\,\mathrm{d}z, \qquad \alpha_f = \operatorname{Re}\omega_f.$$

The modular transformation rule for a weight-two form implies that $\omega_f$ and $\alpha_f$ descend to the quotient. The classical modular-symbol framework identifies such integrals with homological periods on modular curves. [5, pp. 19–25]

Let $X_{\text{geo}}$ be the unit-speed geodesic vector field on $T^1Y_0(11)$. Write $a(v) = \alpha_f(v)$ and set

$$\rho_\varepsilon(v) = 1 + \varepsilon a(v), \qquad X_\varepsilon = \frac{X_{\text{geo}}}{\rho_\varepsilon}.$$

Here $\rho_\varepsilon$ is the time density, or slowness, and $1/\rho_\varepsilon$ is the speed multiplier. Cusp-form decay makes $a$ bounded, so $|\varepsilon| < \|a\|_\infty^{-1}$ gives a positive time change.

**Proposition 2.1** (period variation)**.** *For an oriented closed geodesic $\gamma$ of base length $\ell(\gamma)$,*

$$T_\varepsilon(\gamma) = \ell(\gamma) + \varepsilon I(\gamma), \qquad I(\gamma) = \int_\gamma \alpha_f.$$

*Moreover, $T_\varepsilon(\gamma^r) = rT_\varepsilon(\gamma)$ for every $r \geq 1$.*

*Proof.* Along the original unit-speed orbit, the new time element is $\mathrm{d}t_\varepsilon = \rho_\varepsilon\,\mathrm{d}t$. Integration gives the first formula. An $r$-fold traversal concatenates $r$ copies of the same path, multiplying both arclength and the one-form integral by $r$. □

### 2.2 Owners, orientation, and repetition

A hyperbolic matrix $M \in \Gamma_0(11)$ determines an oriented quotient loop by projecting an oriented segment of its invariant axis from $z$ to $Mz$. We write $I(M)$ for its $\alpha_f$ period. The next statement fixes the owner used throughout the paper.

**Proposition 2.2** (owner laws)**.** *For $C, M \in \Gamma_0(11)$, with $M$ hyperbolic, and every integer $r \geq 1$,*

$$I(CMC^{-1}) = I(M), \qquad I(M^{-1}) = -I(M), \qquad I(M^r) = rI(M).$$

*Thus $I$ belongs to an oriented $\Gamma_0(11)$ conjugacy class; inversion and repetition are separate operations.*

*Proof.* The invariance $C^*\alpha_f = \alpha_f$ sends an axis segment for $M$ to one for $CMC^{-1}$ without changing its integral. Reversing a path changes the sign. A path for $M^r$ is the concatenation of $r$ translates of a path for $M$, and quotient invariance makes the summands equal. □

This distinction is essential. A primitive owner is not replaced by its $r$-fold traversal in the primitive product, while orientation reversal is an independent primitive orbit for the oriented geodesic flow.

## 3 Related work and claim position

Manin's modular-symbol construction places weight-two periods in the homology of modular curves, and Merel gives an explicit treatment of Hecke operators on relative homology using the $\mathbb{P}^1(\mathbb{Z}/N\mathbb{Z})$ model. [6, Introduction and Sections 1–2] These sources supply the homological period object and the Hecke action used here. They do not identify a correspondence branch with one primitive orbit, distinguish branch-cycle degree from product repetition, or assert a scalar primitive-Euler recurrence for the time-changed flow. The present increment at this interface is the explicit cycle-pushforward owner decomposition followed by a degree-wise quadratic-moment test on a frozen output multiset.

Katok's earlier work places closed-geodesic periods and modular-form arithmetic in direct contact [3]. It is an antecedent for the period object, not for the present finite correspondence-component owner taxonomy or degree-moment obstruction.

Ruelle introduced zeta functions organized by primitive periodic orbits of hyperbolic dynamics, and Fried carefully separated prime periodic orbits from their iterates in comparing Ruelle and Selberg products. [7, Introduction and flow case] [2, Section 2] Those works provide the primitive-owner and repetition conventions against which the present bookkeeping is tested. Here the owner family is instead the registered finite correspondence-component multiset, and the repetition series is used only for formal coefficient comparison. No transfer operator, global determinant, convergence theorem, meromorphic continuation, or global primitive-owner census is imported from those analytic theories.

A modern nearest neighbor proves arithmetic-statistical nonvanishing results for automorphic-form periods over primitive closed geodesics [1, Introduction and main theorems]. Its owners are primitive geodesics ordered in an asymptotic family; it does not turn a finite Hecke correspondence output into a canonical primitive-owner Euler product.

The source-verified comparison available in this article is nearest at the level of ingredients rather than an exhaustive map of modern literature. Manin and Merel provide the period and Hecke-homology setting; Ruelle and Fried provide primitive-owner and repetition conventions. Katok supplies an arithmetic closed-geodesic-period antecedent, while Constantinescu–Nordentoft supplies a modern primitive-geodesic nonvanishing comparison. Neither nearest work provides the cycle-pushforward owner decomposition, declared correspondence-component multiplicity, or degree-wise quadratic-moment classification used here. The bounded increment is the combination of those ingredients with an all-parameter degree-moment non-implication, the exact $2/2/134$ taxonomy on 138 registered output instances, and the $51/55$ group failures together with a target-blind two-control audit. The comparison is restricted to the seven verified bibliography entries and the hash-bound audit in `notes/stage4_prime_rev02_nearest_work_audit.md`; it does not establish a global priority claim or complete current-literature coverage. In particular, no newform-specific residue is supported by the two-control panel, and no global result is inferred from the finite computation.

## 4 Hecke cycle ownership

For a prime $p \nmid 11$, choose the left-coset representatives

$$\beta_b = \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} \ (0 \le b < p), \qquad \beta_\infty = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}.$$

The weight-two normalization is

$$T_p f(z) = pf(pz) + \frac{1}{p} \sum_{b=0}^{p-1} f\left(\frac{z+b}{p}\right),$$

equivalently $T_p\omega_f = \sum_\beta \beta^*\omega_f$. Since the level-11 cusp space is one-dimensional and $f$ is normalized, $T_p f = a_p f$.

**Theorem 4.1** (cycle-pushforward eigenperiod)**.** *For every oriented closed cycle $C$ on $Y_0(11)$,*

$$\int_{T_{p,*}C} \omega_f = a_p \int_C \omega_f, \qquad \int_{T_{p,*}C} \alpha_f = a_p \int_C \alpha_f.$$

*If $C$ is represented by a hyperbolic $M \in \Gamma_0(11)$, the left side decomposes into a finite sum of explicitly owned closed geodesics.*

*Proof.* Pair the correspondence with the cycle:

$$\int_{T_{p,*}C} \omega_f = \sum_\beta \int_{\beta C} \omega_f = \int_C \sum_\beta \beta^* \omega_f = \int_C T_p \omega_f = a_p \int_C \omega_f.$$

The coefficients $a_p$ are real, so taking real parts proves the second identity.

For the explicit decomposition, right multiplication by $M$ permutes the representative set. There are unique $\pi(j)$ and $\gamma_j \in \Gamma_0(11)$ with

$$\beta_j M = \gamma_j \beta_{\pi(j)}.$$

For a permutation cycle $O = (j, \pi(j), \ldots, \pi^{d-1}(j))$, iteration gives

$$\delta_O := \beta_j M^d \beta_j^{-1} = \gamma_j \gamma_{\pi(j)} \cdots \gamma_{\pi^{d-1}(j)} \in \Gamma_0(11).$$

The projected branch chain closes and is owned by $\delta_O$. Changing the starting branch cyclically conjugates the product in $\Gamma_0(11)$. Summing over cycles reconstructs $T_{p,*}C$. $\square$

The theorem is sum-valued. It does not send one primitive orbit to one primitive orbit. A further control makes this limitation sharp: because $X_0(11)$ has genus one, $H^1(X_0(11), \mathbb{R})$ is generated by the real and imaginary parts of $\omega_f$, and $T_p$ acts by the real scalar $a_p$ on both. Hence every smooth closed real one-form extending to $X_0(11)$ satisfies the same cycle-period relation. The identity is structurally correct but is not discriminative evidence for a primitive Euler mechanism.

## 4.1 Three indices that must remain distinct

The correspondence introduces three superficially similar integers. The first is the branch-cycle degree $d_O$, the length of an orbit of the permutation $\pi$ on the $p+1$ coset representatives. The branch starting at $j$ returns to the same sheet only after $d_O$ applications of $M$, and its closed owner is $\delta_O = \beta_j M^{d_O} \beta_j^{-1}$. Since the fractional-linear action of $\beta_j$ is an isometry of $\mathbb{H}$, the resulting closed path has length $d_O \ell(M)$. This explains the length multiplication without identifying $\delta_O$ with a power of $M$ inside $\Gamma_0(11)$.

The second integer is the primitive-root exponent of $\delta_O$. If $\delta_O = h^q$ for some $h \in \Gamma_0(11)$ and $q > 1$, then the closed path traverses the primitive $h$-geodesic $q$ times. The frozen certificates give $q = 1$ for every registered output. Thus $d_O > 1$ can coexist with a primitive registered owner: $d_O$ records how a Hecke branch closes, whereas $q$ records repetition of one geodesic in the quotient.

**Lemma 4.2** (completeness of the bounded root search)**.** *Let $A$ be any stored positive-trace hyperbolic matrix in $\Gamma_0(11)$. The certificate's finite search over exponents and integral roots is exhaustive for relations $[A] = [B]^q$ in* $\mathrm{PSL}_2(\mathbb{Z})$ *with $B \in \Gamma_0(11)$ and $q \geq 2$.*

*Proof.* Write $T = \mathrm{tr}(A) > 2$. If $[A] = [B]^q$, choose the central sign of $B$ so that its representative $C$ has integral trace $t = \mathrm{tr}(C) \geq 3$. Since both $A$ and $C^q$ then have positive trace, the projective equality lifts to the exact matrix equality $A = C^q$; the possible central sign creates no additional case. For determinant-one matrices define

$$P_0(t) = 2, \qquad P_1(t) = t, \qquad P_j(t) = tP_{j-1}(t) - P_{j-2}(t).$$

Then $P_q(t) = \mathrm{tr}(C^q) = T$. The minimum positive hyperbolic trace is three, so $P_q(3) \leq T$ is necessary. Because $P_q(3)$ grows strictly with $q$, this gives the recorded finite exponent bound. For each exponent within that bound, $P_q(t)$ is strictly increasing on the integers $t \geq 3$, hence at most one integral root trace is possible.

Cayley–Hamilton gives

$$C^q = U_{q-1}(t/2)C - U_{q-2}(t/2)I,$$

where the displayed coefficients are integers under the standard recurrence. Thus the candidate $C$ is uniquely reconstructed from $A$, $q$, and $t$. The verifier rejects it unless all four entries are integral, $\det C = 1$, the lower-left entry is divisible by 11, and $C^q = A$ exactly. The search performs these checks for every admissible $q$. No candidate root exists for any of the 138 registered matrices, so their certificate exponent is one. This proves completeness only for the stored matrices; it does not enumerate primitive conjugacy classes of $\Gamma_0(11)$. □

The third integer is the logarithmic-product repetition $r$ in equations (1)–(2). It appears only after a primitive owner has been selected and enumerates its iterates. Replacing $d_O$ by $r$, or treating the $d_O$ branches as iterates of the source owner, changes both the owner and the coefficient. Finally, two different permutation cycles can in principle yield conjugate output owners. The finite theorem therefore concerns the registered output *multiset*: each correspondence component carries its declared multiplicity even before global cross-instance conjugacy canonicalization. These distinctions explain why a correct linear equality in homology does not automatically become a primitive-orbit product identity.

The following crosswalk collects the three indices and two finite denominators without identifying any of them:

| Symbol or count | Mathematical owner | Role here | Inference not permitted |
|---|---|---|---|
| $d_O$ | one Hecke permutation cycle | branch closure and length $d_O L(M)$ | traversal power or zeta repetition |
| $q$ | one matrix $\delta_O$ in $\Gamma_0(11)$ | primitive-root exponent | Hecke branch degree or product index |
| $r$ | one already selected primitive owner | formal logarithmic repetition | a new correspondence component |
| 138 | registered cycle-owner instances | three-way exact kernel taxonomy | 138 distinct global primitive classes |
| 55 | registered source-word/prime groups | scalar-law pass or failure | an instance percentage or global census |

Both finite counts retain the declared correspondence-component multiplicity. Neither count is a canonical global primitive-conjugacy denominator.

## 5 Zeta variations and exact moment obligations

Write $\mathcal{O}_{\mathrm{fr}}$ for the frozen finite multiset of registered correspondence-component owners, counted with its declared output multiplicity. In the expressions below, the owner sum is taken over a selected inverse-closed subfamily of $\mathcal{O}_{\mathrm{fr}}$, whereas $r \geq 1$ is an unbounded formal repetition index used only to compare coefficients. Thus the adjective *finite* modifies the owner family, not the repetition series. No function space, transfer operator, trace-class or nuclear property, trace formula, Fredholm determinant, convergence or continuation theorem, or global divisor is constructed.

Let $\gamma^\#$ denote an oriented primitive owner, $L = \ell(\gamma^\#)$, and $I = I(\gamma^\#)$. We use two reciprocal finite formal log products:

$$\log Z_R(s,\varepsilon) = \sum_{\gamma^\#} \sum_{r \geq 1} \frac{e^{-sr(L+\varepsilon I)}}{r}, \tag{1}$$

$$\log Z_S^{\mathrm{fr}}(s,\varepsilon) = \sum_{\gamma^\#} \sum_{r \geq 1} \frac{e^{-sr(L+\varepsilon I)}}{r(1-e^{-rL})}. \tag{2}$$

The second convention freezes the transverse multiplier of the base return map. It is a Selberg-type formal product, not the Selberg zeta function of a deformed metric. Every assertion below is exact for a finite owner family and extends termwise only where an infinite family is absolutely and locally uniformly convergent.

**Proposition 5.1** (parity of variations)**.** *The first variations of* (1) *and* (2) *are*

$$\partial_\varepsilon \log Z_R(s,0) = -s \sum_{\gamma^\#} I \sum_{r\geq 1} e^{-srL},$$

$$\partial_\varepsilon \log Z_S^{\mathrm{fr}}(s,0) = -s \sum_{\gamma^\#} I \sum_{r\geq 1} \frac{e^{-srL}}{1-e^{-rL}}.$$

*If the family is inverse closed, both vanish. For one inverse pair, the second variations are*

$$\partial_\varepsilon^2 \log Z_{R,\mathrm{pair}}(s,0) = 2s^2 I^2 \sum_{r\geq 1} re^{-srL},$$

$$\partial_\varepsilon^2 \log Z_{S,\mathrm{pair}}^{\mathrm{fr}}(s,0) = 2s^2 I^2 \sum_{r\geq 1} \frac{re^{-srL}}{1-e^{-rL}}.$$

*Proof.* Differentiating one repeated term once cancels the repeated-period factor $r$ against the logarithmic coefficient $1/r$. The inverse owner has the same $L$ and period $-I$, so odd terms cancel. The paired summand is

$$\frac{2e^{-srL}\cosh(sr\varepsilon I)}{r}.$$

Its second derivative at zero is $2s^2rI^2e^{-srL}$. The frozen stability denominator is independent of $\varepsilon$ and is carried through unchanged. □

For a source pair $(M,p)$, let $d_O$ be the Hecke permutation-cycle degree and set

$$Q_d(M,p) = \sum_{O:d_O=d} I(\delta_O)^2.$$

Although $\delta_O$ has length $d_OL(M)$, Lemma 4.2 proves that the exact root search is exhaustive for each stored matrix, and every registered $\delta_O$ has certificate exponent one. The audited fields include the exponent bound, the primitive verdict, the reconstructed root exponent, exact determinant and subgroup membership, and the final exact-power check. Hence $d_O$ is not the zeta repetition $r$ of $\delta_O$ on this registered multiset. This finite certificate does not assert a global primitive-class enumeration.

**Theorem 5.2** (quadratic degree-moment criterion)**.** *Let $\lambda_p$ depend only on $p$ and be fixed independently of $M$ and $s$. On the finite Hecke output multiset, the inverse-paired identity*

$$\sum_O V^{(2)}(\delta_O;s) = \lambda_p V^{(2)}(M;s)$$

*holds for every sufficiently large real $s$, for either kernel in Proposition 5.1, if and only if*

$$Q_1(M,p) = \lambda_p I(M)^2, \qquad Q_d(M,p) = 0 \quad (d>1).$$

*Proof.* Put $q=e^{-sL(M)}$. After removing $2s^2$, the Ruelle output has coefficient

$$n\sum_{d|n} \frac{Q_d}{d}$$

at $q^n$, whereas the source coefficient is $n\lambda_pI(M)^2$. Equality on an interval of $q$ gives equality of all Taylor coefficients. Dividing by $n$ and applying Möbius inversion yields $Q_1=\lambda_pI(M)^2$ and $Q_d=0$ for $d>1$. Conversely these conditions give the identity term by term. In the frozen-stability kernel, every divisor contribution at degree $n$ has the same nonzero factor $(1-e^{-nL})^{-1}$, so the same inversion applies. □

The linear relation $\sum_O I(\delta_O) = a_pI(M)$ cannot imply this sum-of-squares condition. The obstruction is algebraic before any root-finding experiment is contemplated.

### 5.1 Why the linear eigenperiod law does not determine the square moments

Assume first that $I(M) \neq 0$ and write

$$u_O = \frac{I(\delta_O)}{I(M)}.$$

The Hecke eigenperiod identity determines the single signed statistic

$$\sum_O u_O = a_p.$$

The second variation instead asks for one nonnegative statistic in every degree bin,

$$\sum_{O:d_O=d} u_O^2 = \begin{cases} \lambda_p, & d = 1, \\ 0, & d > 1. \end{cases}$$

There is no algebraic implication from the first display to the second. Even within one degree bin, replacing a pair $(u_1, u_2)$ by $(u_1 + t, u_2 - t)$ preserves its signed sum while changing the square sum by $2t(u_1 - u_2) + 2t^2$. Across degree bins, the linear relation does not record where any contribution occurs. Because the second moments are sums of rational squares, the condition at a nonunit degree is especially rigid: it holds exactly when every real period in that bin vanishes.

The all-$s$ quantifier is what exposes this rigidity. For a single numerical value of $s$, distinct degrees can compensate after being weighted by different exponentials. Equality for every sufficiently large $s$ is equality of a convergent power series near $q = 0$ and therefore fixes every coefficient. Möbius inversion then removes the zeta repetitions and leaves the branch-degree moments. The exact test does not fit $\lambda_p$ to those coefficients: the three laws were frozen first, and the builder merely evaluates the resulting rational identities.

The case $I(M) = 0$ would require a separate unnormalized formulation, since the ratios $u_O$ would be undefined. It does not occur in the registered source panel: Theorem 6.1 and the source ledger certify $k(M) \neq 0$ for all eleven sources. This nondegeneracy is part of the finite claim boundary, not an assumption about all hyperbolic classes of $\Gamma_0(11)$.

**Corollary 5.3** (nonunit mass is a scalar-law obstruction)**.** *If $Q_d(M, p) > 0$ for any $d > 1$, then no choice of scalar $\lambda_p$ can make the all-$s$ second-variation recurrence hold for that source group. This conclusion is independent of the degree-one moment.*

*Proof.* Theorem 5.2 requires $Q_d = 0$ at every nonunit degree, and $\lambda_p$ appears only in the degree-one equation. Since $Q_d$ is a sum of real squares, a positive value cannot be canceled by another owner or by changing the scalar. □

This corollary separates structural failure from normalization failure. A group that has the correct degree-one square sum but positive mass at degree two or higher is not a near success awaiting a better Hecke polynomial; its length support is wrong. In contrast, a group with no nonunit mass can still fail because its degree-one square sum is not the predeclared $\lambda_p$. The ledger records both flags, which is why the later counts can distinguish the 47 double failures from the four $a_p^2$ groups whose only obstruction is nonunit support.

## 6 Exact Schreier homology classifier

The finite taxonomy uses

$$\mathrm{PSL}_2(\mathbb{Z}) = \langle s, r \mid s^2 = r^3 = 1 \rangle$$

and identifies the twelve right cosets of $\Gamma_0(11)$ with $\mathbb{P}^1(\mathbb{F}_{11})$. A deterministic Schreier transversal produces 24 arcs. The rewritten $s^2$, $r^3$, and spanning-tree relations give a 35-row rational relation matrix of rank 21. Consequently

$$\dim_{\mathbb{Q}} H_1(Y_0(11), \mathbb{Q}) = 3.$$

In the frozen coordinate basis, the cusp direction is $(-1, 0, 0)$; quotienting its span gives the two-dimensional compact homology of $X_0(11)$.

Complex conjugation on the modular curve is induced by $z \mapsto -\overline{z}$. Exact rewriting on a rational basis gives the involution

$$\tau(x, y, z) = (-x, y + z, -z), \qquad h + \tau h = (0, 2y + z, 0).$$

Set $k(x, y, z) = 2y + z$. The compact $+1$ eigenspace is one-dimensional, and real integration of $\omega_f$ factors through it.

**Theorem 6.1** (exact period coordinate)**.** *For every frozen source owner $M$, $k(M) \neq 0$. For every frozen Hecke cycle owner $\delta$,*

$$\frac{\operatorname{Re} \int_\delta \omega_f}{\operatorname{Re} \int_M \omega_f} = \frac{k(\delta)}{k(M)} \in \mathbb{Q}.$$

*The real period vanishes exactly when $k(\delta) = 0$. The full complex period vanishes exactly when the compact homology class of $\delta$ is zero.*

*Proof.* For a rational homology class $h$, invariance of the real part under conjugation gives

$$2 \operatorname{Re} \int_h \omega_f = \int_{h+\tau h} \omega_f.$$

The symmetrized class is $(0, k(h), 0)$ modulo the cusp direction, so all real periods are scalar multiples of the single compact $+1$-eigenperiod. Taking the ratio for $h = [\delta]$ and $[M]$ gives the formula. The builder verifies $k(M) \neq 0$ for all eleven frozen source owners. Finally, a cusp form extends holomorphically to the compactification, so cusp cycles have zero period. On the genus-one compact curve, integration of a nonzero holomorphic differential embeds rational homology in its period lattice; hence zero full complex period is equivalent to zero compact class. $\square$

This theorem replaces numerical near-zero decisions by exact rational algebra. In particular, normalized moments are

$$M_d = \sum_{O:d_O=d} \left( \frac{k(\delta_O)}{k(M)} \right)^2 \in \mathbb{Q}_{\geq 0}.$$

For $d > 1$, $M_d = 0$ if and only if every owner in that bin lies in the exact real-period kernel.

For each registered owner, the exact classification chain is

$$\delta \longmapsto [\delta]_{\text{Sch}} \in \mathbb{Q}^3 \longmapsto \overline{[\delta]} \in H_1(X_0(11), \mathbb{Q}) \longmapsto k(\delta) = 2y + z \longmapsto \text{label}.$$

The first arrow is exact Schreier rewriting of the owner matrix, the second quotients the cusp direction, and the third takes the real-involution coordinate. The final decision has three mutually exclusive rows:

| Exact condition | Label and meaning |
|---|---|
| $\overline{[\delta]} = 0$ | full complex-period kernel; both period components vanish |
| $\overline{[\delta]} \neq 0$ and $k(\delta) = 0$ | real-projection-only kernel; the complex period is nonzero |
| $k(\delta) \neq 0$ | true real-period nonkernel |

This chain classifies only the frozen correspondence-component multiset. It neither canonicalizes conjugacy across output instances nor supplies a global primitive census.

### 6.1 Kernel semantics and exact-decision hierarchy

The two kernel labels answer different questions. A full complex-period kernel has zero class in compact homology, so both the real and imaginary periods of $\omega_f$ vanish. A projection-only kernel has a nonzero compact class but zero $+1$ component under the real involution; its real period vanishes while its complex period does not. Conflating these cases would erase the mechanism behind two of the four exceptional instances. Conversely, a true nonkernel has $k \neq 0$, which is a proof of nonvanishing of the real period, not merely a failure to recognize zero.

The decision hierarchy is intentionally one-way. Matrix identities and congruence membership are checked over the integers. Schreier rewriting and row reduction are performed over $\mathbb{Q}$. Compact-class and $k$ tests are therefore exact equalities. Only after an exact status has been assigned does the verifier compare inherited binary64 period data as a diagnostic. A small numerical residual can never create a kernel row, and a larger residual can never overturn one. Missing coordinates, failed owner reconstruction, or a broken input hash produce a fail-closed error rather than a fourth empirical category.

This hierarchy also clarifies what the enumeration proves. Exhausting 138 locked rows proves a statement about every row of that finite population because the population hash and row count are inputs to the certificate. It does not prove that the population exhausts all primitive conjugacy classes, all positive words, or all Hecke images. The global and finite quantifiers remain explicit throughout.

## 7 Complete finite taxonomy

The input population was frozen before classification: eleven positive-word source owners, five primes $p \in \{2, 3, 5, 7, 13\}$, 55 word/prime groups, and 138 Hecke permutation-cycle owner instances. The tested laws were $\lambda_p = a_p$, $\lambda_p = a_p^2$, and the predeclared secondary control $\lambda_p = a_p^2 - p$. No group or scalar was selected after viewing the exact taxonomy.

**Theorem 7.1** (exhaustive frozen taxonomy)**.** *The 138 owner instances split mutually exclusively into two full complex-period kernels, two nonzero real-projection-only kernels, and 134 true real-projection nonkernels. There are no degenerate or unresolved instances. The per-prime counts are given in Table 1.*

Table 1: Exact classification of the frozen Hecke cycle-owner instances.

| $p$ | full kernel | projection-only | true nonkernel | total |
|---|---|---|---|---|
| 2 | 0 | 0 | 18 | 18 |
| 3 | 0 | 0 | 22 | 22 |
| 5 | 2 | 2 | 26 | 30 |
| 7 | 0 | 0 | 30 | 30 |
| 13 | 0 | 0 | 38 | 38 |
| All | 2 | 2 | 134 | 138 |

*Proof.* For every locked cycle row, the verifier reconstructs $\delta_O$ from $\beta_j M^{d_O} \beta_j^{-1}$, checks exact equality with the stored matrix, determinant one, the lower-left congruence modulo 11, and the finite primitivity certificate. It then converts the matrix into an $s, r$ word, applies exact Schreier rewriting over `fractions.Fraction`, and computes its compact class, conjugate class, and $k$ coordinate. The mutually exclusive decision rule is: compact class zero gives a full kernel; compact class nonzero and $k = 0$ gives a projection-only kernel; $k \neq 0$ gives a true nonkernel; unavailable exact data would fail closed as degenerate. Exhaustive iteration produces the displayed counts and zero degenerate rows. This is a finite proof by exact enumeration; the checked program and data hashes are part of the certificate described in Section 8. □

The finite primitivity step in this proof is complete for every stored matrix by Lemma 4.2. The Round-4 ledger records `root_search_max_exponent`, `primitive_in_gamma0_11_exact`, and `primitive_root_exponent`; the Round-8 instance ledger independently records the recomputed primitive verdict and exponent. All 138 rows have exponent one. These fields certify only the registered instances and do not deduplicate or enumerate global conjugacy classes.

The four kernels all occur at $p = 5$ and cycle degree five. The words LRRLRRR and LLRLLRLR have compact-zero output classes. The words LLLRLLRLR and LLLRLRLLR have nonzero compact classes that are anti-invariant under $\tau$, so their complex periods are nonzero and purely imaginary while their real projections vanish.

**Corollary 7.2** (group-level recurrence verdicts)**.** *For each primary law $a_p$ and $a_p^2$, exactly four of 55 groups satisfy all quadratic degree moments and 51 fail. The four survivors are the $p = 5$ kernel groups above. The secondary control $a_p^2 - p$ fails all 55 groups.*

*Proof.* For every group and law, the verifier forms the exact rational square sums $M_d$ and subtracts the required moment $\lambda_p$ at degree one and zero at every nonunit degree. The two primary laws coincide at $p = 5$ because $a_5 = 1$. Exactly the four kernel groups have $M_1 = 1$ and $M_d = 0$ for $d > 1$. For $a_p$, all 51 failures violate both the degree-one condition and a nonunit condition. For $a_p^2$, 47 violate both and four satisfy degree one but retain nonunit mass. The control changes the required degree-one scalar and therefore rejects even the four kernel groups. □

The exact mechanisms separate the primary failures more sharply than a success percentage. For $a_p$, all 51 studied-coordinate failures violate both the degree-one scalar condition and a nonunit-degree condition. For $a_p^2$, 47 violate both conditions and four satisfy degree one but retain nonunit mass. In the matched-control audit, both controls fail on 51, 44, and 55 of the studied-coordinate failures for $a_p$, $a_p^2$, and $a_p^2 - p$, respectively; the remaining seven $a_p^2$ failures fail exactly one control, and none of the studied-coordinate failures passes both controls. The diagnostic flags *degree-one absent with nonzero scalar* and *negative scalar* overlap and are not an additive partition: their counts are 17 for each law and 33, 0, and 33 across the three laws, respectively. These are finite exact diagnostics, not estimates of a population beyond the registered groups.

There are two denominators in these statements. The instance denominator 138 counts individual permutation-cycle components and supports the three-way kernel taxonomy. The group denominator 55 counts source-word/prime pairs and is the unit on which a scalar recurrence is either satisfied or rejected. One exceptional instance can affect its whole group, while several instances in a group contribute to the same degree moment. Reporting only an instance success percentage would therefore answer the wrong question. Conversely, reporting only 51 failed groups would hide why the four surviving groups are non-generic. The paired ledgers preserve both levels of ownership.

Both denominators are correspondence-component quantities counted with the declared finite multiplicity. The branch degree $d_O$, primitive-root exponent $q$, and formal repetition $r$ act within these levels as shown in the crosswalk above; none turns either denominator into a canonical global primitive-owner census. Cross-instance owner canonicalization and any recomputation under a deduplicated multiplicity rule remain outside this result.

Nor are the four studied-coordinate survivors validation data for a revised law. All occur at the single prime $p = 5$, where the two primary scalars coincide, and all are explained by exact kernel geometry. The target-blind controls $y - z$ and $y - 2z$ produce different individual pass patterns, showing that a finite verdict can depend on the selected cohomology functional; nevertheless, no failure for $k = 2y + z$ is accompanied by passes for both controls. Under the frozen protocol, the four groups remain classified positives for the tested finite identity while functioning as adversarial controls against an arithmetic interpretation. They do not validate a new law, establish newform uniqueness, or support extension to an unregistered source or prime.

# 8 Certificate and reproducibility method

The Round-8 builder is a Python-standard-library program. Its locked inputs are

- `round4_hecke_cycle_ledger.csv`, with SHA-256 `f906df349b8f1fa2864fed592792e0ff f63ba246a069179b7bd8cfdf46520662`; and
- `round6_quadratic_degree_moment_ledger.csv`, with SHA-256 `f95e1435c9293f8e008cebf80084ea2b 522b76186dbd684b5e3997c5e588edea`.

Exact rational homology and rational square sums decide the theorem; inherited binary64 periods are retained only as cross-checks.

The generated instance ledger has 138 rows, and the group/law ledger has 165 rows. All 165 exact verdicts agree with the earlier numerical statuses, with maximum normalized cross-check residual $1.9895196601282805 \times 10^{-13}$. This agreement is not used as proof. Eighteen Round-8 unit tests cover input hashes, population counts, the Schreier model, the involution formula on every owner, exact owner rebuilding, primitivity, taxonomy counts, square-sum identities, law verdicts, fail-closed tampering, route boundaries, and deterministic in-memory rebuilding.

The separately bound Stage-4-prime support layer adds 10 dependency-closure, artifact-binding, matched-control, tampering, and no-refresh checks; all 10 passed in the verify-only replay. These checks audit provenance for the existing scientific values rather than create new values.

A separate Stage-4 support suite adds ten direct tests for transitive dependency closure, fail-closed hash drift, deterministic matched-control selection and serialization, population preservation, exact decomposition, and canonical-byte binding. These ten tests are supplemental and are not folded into the registered count of eighteen Round-8 unit tests. The root-certificate surface remains row-auditable through the Round-4 exponent-bound, primitive-verdict, and root-exponent fields and the corresponding Round-8 recomputed verdict and exponent for all 138 instances.

Running `experiments/reproduce_round8.sh` performs the tests, generates two isolated four-file trees, compares them byte for byte, and checks the result against the repository artifacts. The two tree hashes are both

`cc36c1f952c9ce89050996f4bb4c9905571f9ef09a0d7115be8a985e02a5621d`.

The computation is therefore a reproducible exact certificate for the registered finite population, not a numerical sample used to infer an unbounded theorem.

The support replay command is `bash experiments/reproduce_stage4_round8_support.sh`. It also produced two byte-identical support trees with SHA-256 `c3a254b0834b217b074eb5d700a13f805853634578ae0ce7d9fbdd64280e85fd` and verified that the four canonical Round-8 result files retained their committed bytes.

The supplemental command `experiments/reproduce_stage4_round8_support.sh` first replays that legacy certificate and then builds two isolated support trees containing the dependency manifest and matched-control outputs. The two supplemental tree hashes are both `c3a254b0834b217b074eb5d700a13f805853634578ae0ce7d9fbdd64280e85fd`. The legacy four-file tree and its displayed SHA-256 remain unchanged, and the support verifier rechecks the canonical bytes before reporting success.

The certificate is layered so that each conclusion can be audited at its natural level. The source lock prevents a later change of words, primes, or candidate laws. Owner rebuilding verifies that each stored matrix really arises from the declared source, coset branch, and permutation cycle. The homology layer derives, rather than imports, the compact class and involution coordinate. The moment layer groups exact ratios by branch degree and evaluates all three predeclared laws. Finally, the serializer sorts rows canonically and hashes the complete output tree. Agreement of two isolated trees checks determinism; agreement with the checked-in tree checks repository integrity. Neither check replaces the mathematical proof, but together they bind the proof's finite hypotheses to inspectable artifacts.

The transitive project-source closure is made explicit by `notes/stage4_round8_dependency_manifest.json` (SHA-256 `04544ef268f8b41e526b013bc77d52bb6faf96495e0b88e435dc0d6dd7f3e0dc`; closure SHA-256 `bafd36828ebe11ca6c467f9b26c8902170f30b6d75f87ff801de73764d44523e`). This closed graph is the full support scope asserted here; it is not a proof of PDF byte reproducibility or of an unbounded dynamical claim.

The supplemental dependency manifest closes the transitive project-source graph

$$\texttt{round8} \longrightarrow \texttt{round7} \longrightarrow \{\texttt{round2}, \texttt{round4}\}, \qquad \texttt{round4} \longrightarrow \texttt{round2},$$

and binds all four source files by byte count and SHA-256. It supplements rather than rewrites the legacy Round-8 manifest. At the primitivity layer, the locked Round-4 ledger carries the finite exponent bound and subgroup-root checks, while the Round-8 instance ledger recomputes the primitive verdict and exponent for each of the 138 registered matrices. Thus both dependency closure and bounded root completeness are inspectable without changing the canonical result tree.

Several common failure modes are therefore observable. Changing one input byte breaks the source hash. Dropping or duplicating a row breaks population invariants. Substituting floating-point zero tests changes the schema expected by the validator. Reclassifying a projection-only kernel as a full kernel breaks the compact-class consistency check. Editing a Route verdict without changing the underlying evidence breaks the boundary assertions. The test suite includes representative tampering cases, so reproducibility means more than rerunning a script that always exits successfully.

The Stage-4-prime support tests exercise those bindings and fail closed on representative source, result, and Route tampering. Their receipt, `experiments/stage4_round8_support_receipt.json` (SHA-256 `70f3b2acabeca6327a98cd4fe5f5b8910b104a1cc6a4171fe7dea91997cf7c10`), records 18 legacy tests and 10 support tests passing with zero failures and canonical result bytes unchanged.

The supplemental regression suite also rejects an altered transitive-source hash and a manifest that omits a required project dependency. Its refresh mode writes only the separately named Stage-4 support artifacts; verify-default mode has no canonical write path. Before and after both isolated runs, the verifier checks the hashes and byte counts of the legacy manifest, ledgers, summary, and receipt. A dependency omission, scientific-value change, population change, or Route-boundary change therefore stops the support replay rather than refreshing a canonical result.

## 9 Adversarial controls and Route-A interpretation

The control analysis separates the five Route-A layers. At A0, the level-11 newform supplies genuine arithmetic structure, and no prime table or Riemann-zero table enters the object. Nevertheless, there is no intrinsic rational-prime owner map or natural $\log p$ clock, so the arithmetic relation remains weak. At A1, the period owner and finite Hecke cycle owners are explicit and reproducible, but the positive-word source ledger is not a complete primitive $\Gamma_0(11)$ conjugacy census. The appropriate verdict is therefore weak rather than a certified full primitive-orbit layer.

At A2, no global dynamical determinant was built, no training/validation/test divisor regions were defined, and no argument-principle root count, missing/extra-zero report, cutoff drift, or precision drift was run. The finite log-product obstruction does not substitute for those requirements. A3 fails because no functional equation, gamma factor, analytic continuation, counting law, or Weil-type compression was derived from this candidate. A4 fails because no natural unitary or scattering lift preserving this clock and these weights was constructed. The conservative tuple is

$$(\text{A0\_WEAK\_ARITHMETIC\_RELATION}, \text{A1\_WEAK}, \text{A2\_FAIL}, \text{A3\_FAIL}, \text{A4\_FAIL}),$$

with overall status `ROUTE_A_EXPLORATORY`; Route B is not authorized.

Two structural controls remain important: every compactly extending closed real one-form on the genus-one curve obeys the same scalar Hecke cycle relation, and inverse pairing cancels every first variation while turning each second variation into a square. To make that observation quantitative without fitting target labels, the Stage-4 audit enumerated primitive integer coefficient pairs in $(y, z)$ by increasing $\ell^1$ norm and lexicographic order, fixed a canonical sign, excluded the studied direction $(2, 1)$, and retained the first two independent pairs nonzero on all eleven frozen sources. This target-blind rule selected $y - z$ and $y - 2z$ using only frozen source words and exact Schreier source coordinates; it used no output classification, scalar-law verdict, target data, formal A2 result, determinant, or Route-B input.

Among failures for $k = 2y + z$, both matched controls fail in 51 groups for $a_p$, 44 for $a_p^2$, and 55 for $a_p^2 - p$; seven $a_p^2$ failures fail exactly one matched control, and no studied failure passes both. The predeclared $a_p^2 - p$ law still rejects all 55 studied-coordinate groups. This supports a generic finite obstruction only relative to the declared two-control panel. It is not a theorem over every closed one-form, a proof of newform uniqueness, or evidence for a determinant, A2 promotion, or Route B.

# 10 Limitations and open obligations

The taxonomy is complete only for the frozen output multiset. It does not deduplicate $\Gamma_0(11)$-conjugate owners across different output instances, enlarge the source word cutoff, or enumerate all primitive conjugacy classes. Although each of the 138 stored instances has an exact finite root certificate, this does not show that there are 138 globally distinct owners. Cross-instance conjugacy canonicalization is the smallest next finite task.

Accordingly, every primitive-Euler statement in this article uses the registered correspondence-component multiset and its declared finite multiplicity as the owner domain. A product indexed by globally canonical primitive owners, together with its multiplicity rule, would require the separately authorized cross-instance canonicalization and recomputation that have not been run here.

The analytic scope is also local. The formal products are differentiated on finite registered families of correspondence components with their declared multiplicity; they are not claimed to be a canonical primitive-owner Euler product. No convergence or continuation theorem is proved for the full cusped time-changed flow. A different, genuinely non-scalar Hecke action on a transfer operator is not rejected by Theorem 5.2. The theorem refutes only the registered $p$-only scalar recurrences. Its evidence domain is the registered 138-instance/55-group multiset. Finally, the study provides no rational-prime dictionary, target-zero comparison, or quantum operator. These omissions prevent promotion to Route-A A2 and prevent any Route-B inference.

# 11 Conclusion

The level-11 newform time change has a clean arithmetic period coordinate and an exact Hecke relation, but the relation is owned by a cycle pushforward rather than one primitive orbit. Once branch-cycle degree, root exponent, formal repetition, orientation, and declared correspondence-component multiplicity are kept distinct, the tested primitive-Euler interpretation fails on the registered 138-instance/55-group multiset. Exact Schreier homology gives the $2/2/134$ finite taxonomy, and the two primary laws each fail 51 groups. The target-blind $y - z$ and $y - 2z$ audit finds no studied-coordinate failure for which both controls pass, so it supports only a generic finite obstruction and supports no newform-specific residue within that panel.

This finite owner-and-moment conclusion is distinct from existence or asymptotic nonvanishing results for closed-geodesic periods [1, 3]. The bounded nearest-work audit supports that distinction without asserting exhaustive coverage, global priority, or a complete primitive census.

The source-verified contribution comparison remains bounded to the period, Hecke-homology, and classical primitive-product sources cited here; it is not an exhaustive modern-priority claim. The paper-level contribution is therefore a finite owner-level non-implication theorem, an exact taxonomy, and an auditable negative-control decomposition. It is not a canonical global primitive-owner census, a determinant construction, or a Hilbert–Pólya candidate. The formal Route-A tuple remains

`(A0_WEAK_ARITHMETIC_RELATION,A1_WEAK,A2_FAIL,A3_FAIL,A4_FAIL)`, and Route B remains uninvoked.

## Data and Code Availability

All canonical data used in this study remain the repository artifacts under `papers/26-level11-newform-time-change/results`. The deterministic Round-8 source, eighteen-unit-test suite, and verify-default reproduction script remain under `code` and `experiments`.

Stage-4-prime provenance additionally binds `notes/stage4_round8_dependency_manifest.json`, `code/stage4_round8_support.py`, `code/test_stage4_round8_support.py`, `experiments/reproduce_stage4_round8_support.sh`, `experiments/stage4_round8_support_receipt.json`, and the two `stage4_matched_exact_control` result files. The support command is verify-default and does not refresh the canonical Round-8 results.

The supplemental Stage-4 dependency closure is `notes/stage4_round8_dependency_manifest.json` with SHA-256 `04544ef268f8b41e526b013bc77d52bb6faf96495e0b88e435dc0d6dd7f3e0dc`. The matched exact decomposition is `results/stage4_matched_exact_control_decomposition.csv` with SHA-256 `6681e1c34f260b7b29ccfd7b24c7d31c204036a7a481e74fc3d40fff9e507e9a`. The matched exact summary is `results/stage4_matched_exact_control_summary.json` with SHA-256 `bcb6484321a6064100e8e7ed5b7dd64d75416a31e8097115afc41022d7f3991f`. The support receipt is `experiments/stage4_round8_support_receipt.json` with SHA-256 `70f3b2acabeca6327a98cd4fe5f5b8910b104a1cc6a4171fe7dea91997cf7c10`. These supplemental artifacts bind the unchanged canonical bytes and require no proprietary or target-side dataset.

## Ethics Declaration

This theoretical and computational mathematics study involved no human participants, personal data, animals, clinical material, or field intervention. Institutional ethics approval and informed consent were therefore not applicable.

## Author Contributions

Liang Wang: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing–Original Draft, Writing–Review & Editing, Visualization, and Project Administration.

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The author declares no conflict of interest.

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AI-assisted tools were used to support research organization, source-metadata checking, code development, mathematical exposition, and manuscript drafting. The author retains responsibility for the definitions, theorem statements, proofs, computations, citations, claim boundaries, and final text. Exact computational claims are accompanied by deterministic tests, source locks, and reproducibility receipts; AI output is not treated as independent evidence.

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